

Agile Team Charter for SE4450

Team identity

Guidelines: State who the team is, what project is being delivered, and the outcomes expected this term to align assessment and stakeholder goals.

- Team name and logo: **Ctrl + Incision**



- Project title: *Enhancement and verification of Python-based software for image-guided surgery.*
- Project Mission (1–2 sentences): *Our goal is to improve and test the existing Python-based software for image-guided surgery so that it is modular, easy to extend, and reliable in real time. We will also investigate methods for reconstructing 3D volumes from sequential 2D ultrasound images utilizing tracking data, and re-implement key C++ components in Python to streamline the system architecture while preserving performance and accuracy.*

Roles and responsibilities

Guidelines: Clarify who holds which responsibilities in this course context. If no industry sponsor is available, the instructor assumes Sponsor duties; the Faculty Advisor mentors/grades the team's technical and academic rigor.

- Instructor (Course Lead): *Abdelkader Ouda, aouda@uwo.ca* — oversight, grading rubric alignment, milestone governance, cross-team arbitration; sponsor role if no external partner.
- Faculty Advisor (Team Coach): *Elvis Chen, chene@robarts.ca* — technical mentorship, scope realism, regular check-ins, co-signs key deliverables per program policy.
- Industry Sponsor (if applicable) or Instructor as Sponsor: *Abdelkader Ouda, Western University / aouda@uwo.ca* — problem context, success criteria, timely feedback, data/assets, milestone reviews.

- Product Owner (student): *Elise Drouillard* — orders backlog, writes acceptance criteria, schedules review, and integrates feedback.
- Scrum Master/Facilitator (student): *Hannah Taylor* — facilitates ceremonies, removes impediments, tracks improvement actions.
- Developers/Engineers (students):
 - *Elise Drouillard* / UI/UX design, API development, modularity
 - *Janki Patel* / UI/UX design, system modularity, QA and testing
 - *Hannah Taylor* / UI/UX design, system modularity, prototyping
 - *Delaney McLachlan* / 3D imaging, Test Case Writing
 - *Muneeb Asif* / UI/UX design, 3D imaging reconstruction, algorithm development, prototyping, pipeline integration

— cross-functional delivery (UI/API/data/QA/UX), own quality and DoD.

- Stakeholders/SMEs (if any): *N/A* — feedback cadences, decisions required.
- A draft of a RACI (Responsible, Accountable, Consulted, and Informed) chart, which will be updated per project phases/sprints. Provide a link to the repo.

Task	Elise	Hannah	Janki	Delaney	Muneeb
System architecture review (Python modules, dependencies, integration points)	A	I	R	C	C
Code enhancement (new features, bug fixes, refactoring)	R	C	C	I	I
Verification planning (test plan, acceptance criteria)	I	C	R/A	R	C
Unit and integration testing (Python test scripts, simulation data)	I	I	C	C	R
Validation against clinical use cases (mock data, simulated workflow)	R/A	I	I	C	I
Documentation (technical design, user instructions, version control)	R	R/A	C	I	C
Quality assurance & compliance checks (coding standards, reproducibility)	C	C	C	R	I
Demo preparation & presentation (slides, software walkthrough)	C	R	C	R	C
Repository management (branching, pull requests, merges)	I	R	R	C	A

- **R (Responsible):** Does the work
- **A (Accountable):** Ultimately answerable for the task's completion
- **C (Consulted):** Provides input, subject matter expertise
- **I (Informed):** Kept updated on progress

Each task/row can have multiple people assigned to one role.

Working agreements

- **Pull Requests:** All PRs are reviewed within 24 hours; pair review for complex or large changes.
- **Coding Standards:** Follow shared style guide (documented in repo); enforce with tests.
- **Availability:** Publish working hours/availability in a shared calendar.
- **Core Hours:** Aim to overlap at least 2–3 hours/day for synchronous collaboration.
- **Communication:** Use project group chat as primary; reply within 24h.
- **Meetings:** Start/end on time; agenda shared in advance; decisions documented.
- **Retro Actions:** Commit to 1–2 improvements per sprint with owners/dates; verify next sprint.
- **Issue Tracking:** All work tracked in GitHub Issues/Project Board; no side work off the record.

Decision rules

Routine Decisions (two-way doors):

- The implementer decides (e.g., function naming, minor refactor).
- Must document in PR notes or issue comments.

Significant Design/Scope (one-way doors):

- Require an Architecture Decision Record (ADR) or written note in the repo.
- Decision by consent among the *Product Owner (student)* and the *Faculty Advisor*.
- Sponsor input required if requirements/timeline change.

Conflict Escalation:

- Peer mediation first → Faculty Advisor → Instructor.

Timebox Expiry Default:

- If consensus is not reached by the deadline, the team chooses the smallest reversible option and documents rationale.

Communication norms

- System of record for work/decisions: *Github repository (issues, documentation, pull requests)*

- Quick async: *Team iMessage group (primary) and email correspondence for further updates. Important decisions are summarized and documented in GitHub.*
- Formal notices/deliverables: *Email to Instructor (Dr.Abdelkader Ouda) and Faculty Advisor (Dr.Elvis Chen); OWL submission for official deliverables.*
- Cadences:
 - *Stand-up check-ins: 15 min 3x/week via iMessage/email*
 - *Weekly sprint planning, refinement, review, and retrospective (virtual or in-person)*
 - *Advisor/Sponsor (Dr.Elvis Chen) - check-in as scheduled*

Definition of Done (DoD)

Guidelines: Make DoD verifiable and tied to the increment and demo; adjust risk-based checks as needed.

An item is done when:

- Code/feature has been tested against the acceptance criteria.
- Code has passed all tests.
- Code merged to main after all checks passed.
- Docs updated: API/README/changelog; user notes if user-facing.
- Demonstrable in the Sprint Review; included in the potentially shippable increment.

Signatures and acknowledgments

Guidelines: Capture agreement and accountability across academic and sponsor roles.

- Students
 - Elise Drouillard /  / September 17th, 2025
 - Muneeb Asif /  / September 17th, 2025
 - Janki Patel /  / September 17th, 2025
 - Hannah Taylor /  / September 17th, 2025
 - Delaney McLachlan / DM / September 17th, 2025
- Product Owner
 - Elise Drouillard /  / September 17th, 2025
- Scrum Master/Facilitator
 - Hannah Taylor /  / September 17th, 2025
- Faculty Advisor
 - 
Dr. Elvis Chen,
- Industry Sponsor (if applicable) or Instructor as Sponsor
 - Professor Abdelkader Ouda
- Instructor (course lead oversight)
 - Professor Abdelkader Ouda